Entity Graph Alignment and Visual Reasoning for Multimodal Fake News Detection

Guoyi Li
Institute of Information Engineering, Chinese Academy of Sciences
Beijing, China
liguoyi@iie.ac.cn

Die Hu
Institute of Information Engineering, Chinese Academy of Sciences
Beijing, China
hudie@iie.ac.cn

Xiaomeng Fu
Institute of Information Engineering, Chinese Academy of Sciences
Beijing, China
fuxiaomeng@iie.ac.cn

Qirui Tang
Institute of Information Engineering, Chinese Academy of Sciences
Beijing, China
tangqirui@iie.ac.cn

Yulei Wu
University of Bristol, Department of Electrical and Electronic Engineering
Bristol, UK
y.l.wu@bristol.ac.uk

Xiaodan Zhang*
Institute of Information Engineering, Chinese Academy of Sciences
Beijing, China
zhangxiaodan@iie.ac.cn

Honglei Lyu
Institute of Information Engineering, Chinese Academy of Sciences
Beijing, China
lvhonglei@iie.ac.cn

Abstract

The rise of multimodal fake news threatens reliable information dissemination by exploiting multiple modalities to create deceptive, engaging content, significantly impacting society safety. Existing methods still face challenges in cross-modal alignment (e.g., semantic inconsistencies, complex visual-semantic relations) and are vulnerable to low-quality or noisy samples. To address these, we propose Cross-Modal Alignment with Visual Reasoning Prompting (CMA-VRP) for multimodal fake news detection. Specifically, we model text and image entities with graphs to capture fine-grained semantic interactions and enhance cross-modal consistency through graph contrastive learning. Unlike methods relying on shallow image features (e.g., edges, textures), we leverage large language models (LLMs) and large vision-language models (LVLMs) to capture deep visual-semantic attributes related to reasoning (e.g., actions, scenes). Based on graph modeling and visual reasoning features, we perform graph-based cross-modal semantic fusion to unify textual and visual representations and cross-modal cycle alignment to align modality distributions by reducing semantic discrepancies, filtering modality-specific noise, and extracting invariant representations across domains. These steps enable the model to obtain semantically consistent and modality-invariant features. Extensive experiments demonstrate that our model outperforms existing methods in multimodal fake news detection and shows strong robustness against noisy samples.

CCS Concepts

• **Security and privacy** → **Social aspects of security and privacy**; • **Information systems** → *Multimedia information systems.*

*Corresponding author

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Keywords

Multimodal Fake News Detection, Chain-of-thought Prompting, Graph Neural Networks, Contrastive Learning

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1 Introduction

Current methods for fake news detection based on cross-modal alignment and fusion have made significant progress in identifying interactions between text and images during the detection process. By training in relevant domains, these methods enhance understanding of image-text pairs through cross-modal alignment, assessing the consistency between text and images [5, 30, 33, 51, 54]. Despite these advancements, challenges remain in handling multimodal fake news, particularly due to insufficient deep visual semantic modeling capabilities (such as actions, scenes, and object interactions), which leads to the neglect of detailed visual and semantic information in the image, affecting the alignment between visual content and text, and thereby impacting the model's judgment. One reason is that traditional vision-based models typically rely on shallow visual features (e.g., edges and textures) extracted using VGG [42] or ResNet50 [57], overlooking the important role of deep visual features (such as spatial relationships and semantic interactions) in understanding news content. For example, as shown in Figure 1 (a), while the text and image scene appear superficially consistent, the expressions and actions in the image subtly imply violent tendencies, contradicting the text's semantics.

Recently, Large Vision-Language Models (LVLM), such as the BLIP series [23, 24], LLaVA series [26, 27], leverage strong visual instance recognition capabilities and rich world knowledge to assess the consistency of fake news with reality. However, they are less sensitive to subtle cross-modal inconsistencies detection [31, 40], such as discrepancies in actions or slogan content. In fake news,



Figure 1: Examples highlight current challenges. (a) and (b) show the semantic inconsistencies of different modalities in fake news, while (c) and (d) illustrate noise in social media news.

such inconsistencies in details are critical for verification. For example, in Figure 1 (b), if the visual detail of a sign reading "not in my name" held up by an audience member is overlooked, the text and image may mistakenly be considered semantically consistent, leading to an incorrect judgment.

Meanwhile, the rise of fast-paced streaming fuels casual news generation, increasing low-quality content on social media [33, 43, 50]. News images, due to compression or low resolution, may fail to present complete semantic information, often only enabling the recognition of partial entities or reasoning cues (as shown in Figure 1 (c), where only visual entities such as "police car" can be easily distinguished). Some platforms extensively use symbols like "#" for topic tagging or user mentions, or publishers may add prominent title watermarks using image enhancement techniques, which may even carry strong subjective bias (as seen in Figure 1 (d), where content is obscured by unrelated slogan watermarks). These inherent noises, which are unrelated to the news itself, interfere with the model's cognitive ability, thereby affecting the accuracy of detection. This places higher demands on detection models, requiring stronger anti-interference capabilities and the ability to extract reliable semantic information from localized key cues to perform reasoning.

In short, existing multimodal fake news detection faces two main challenges: (1) The lack of comprehensive semantic modeling, especially visual semantics (e.g., actions, scenes, and object interactions), hampers capturing subtle semantic differences, leading to misjudgments and weakening fake news detection, while complicating reliable semantic extraction for reasoning from key cues. (2) The presence of inherent noise in low-quality news [33, 43, 50] (e.g., misleading hashtags, compressed images or intrusive watermarks) disrupts semantic feature extraction, requiring robust reasoning to extract valid information from noise and ensure accurate multimodal alignment and fake news detection.

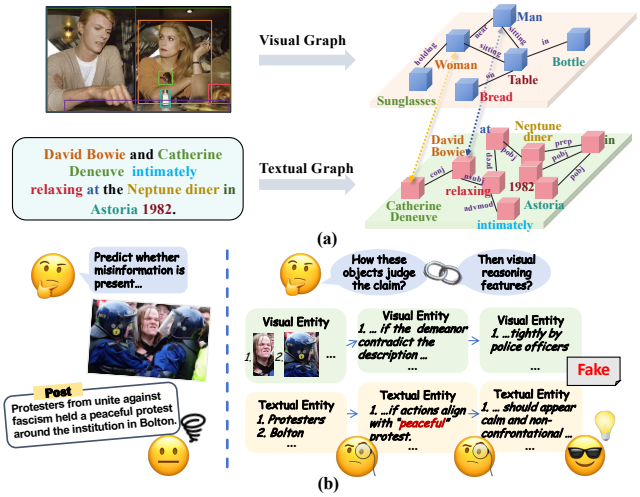


Figure 2: The illustration of the motivation.

Our motivation lies in: (1) Graph modeling detects entity-level inconsistencies in fake news, capturing cross-modal interactions (e.g., actions, scenes, object relationships), as shown in Figure 2 (a). This fine-grained entity-level analysis improves semantic modeling, enhancing reasoning to capture subtle differences [11, 39, 60]; (2) LLMs and LVLMs excel in visual knowledge, reasoning, and entity extraction, offering rich visual reasoning information [13, 15, 35, 37]. Figure 2 (b) shows that capturing semantic differences in cross-modal visual features (e.g., red/underlined text) improves detection. We define visual reasoning features as critical visual semantics aiding claim verification, with LLMs and LVLMs improving the modeling of hidden semantic conflicts and mitigating biases from insufficient deep visual semantics. (3) Existing methods rely on learning explicit alignment features from paired text-image data [30, 60], overlooking the potential of mismatched text-image pairs or data where entities are not explicitly aligned. We propose modality-domain transformation learning to project entities and descriptions into a shared semantic space, enabling the extraction of stable cross-modal associations even from unaligned or noisy data. This alignment removes irrelevant information, reduces noise, and enhances adaptability to low-quality data, improving robustness for reliable fake news detection.

In this paper, we propose Cross-Modal Alignment with Visual Reasoning Prompting (CMA-VRP) for multimodal fake news detection. We construct multimodal entity graphs from news entities and obtain visual reasoning features from LLMs and LVLMs via truth-driven prompting. Given the crucial role of contrastive learning in multimodal fake news representation alignment [16, 51, 61], we introduce entity graph contrastive learning and graph fusion to derive semantically consistent news features by integrating multimodal entity graphs and visual reasoning features. Additionally, modality-domain transformation alignment learns cross-modal semantic mappings, mitigating semantic drift from modality-specific noise and uncovering modality-invariant features, ensuring accurate detection an robust representation learning.

This paper makes several major contributions: (1) We propose a cross-modal alignment method combining multimodal entity graph modeling and visual reasoning fusion. By constructing entity graphs

for textual and visual entities and extracting deep visual semantics, we capture coherent semantics within and across modalities, enhancing deep semantic understanding of news content. (2) Our graph contrastive learning and fusion unify multimodal semantic alignment and deep features, improving cross-modal consistency. (3) We design a CycleGAN-based modality-domain transformation mechanism with visual reasoning features. Learning cross-domain mappings, we mitigate noise and extract modality-invariant features, enhancing model robustness. (4) Extensive evaluation on three real-world datasets demonstrates our model's superior efficiency and robustness.

2 Related Work

2.1 Multimodal Fake News Detection

Existing multimodal fake news detection methods leverage latent information from images and text to construct comprehensive multimodal news representations. Cross-modal alignment plays a crucial role in detecting modality inconsistencies and identifying fake news through semantic relationships [4, 10, 45, 51]. For instance, CAFE [4] integrates cross-modal alignment and ambiguity learning to adaptively fuse unimodal features, while COOLANT [51] employs cross-modal contrastive learning to align image and text features into a unified space, mitigating semantic discrepancies.

2.2 Entity-Enhanced Fake News Detection

Entity alignment is a key approach in fake news detection, leveraging entities' semantic associations, knowledge background, and interactions to reveal latent inconsistencies in multimodal news, enhancing detection [11, 25, 33, 39, 59, 60]. For instance, InfoSurgeon [11] employs entity linking and knowledge graph reasoning to identify semantic inconsistencies between text and images, while EMFEND [39] incorporates visual entity recognition to enhance entity alignment across text and images, improving fake news detection. However, these methods often overlook fine-grained inconsistency modeling and deep alignment, limiting their ability to handle complex semantic scenarios and subtle inconsistencies in fake news. When faced with low-quality data or propagation noise, this issue worsens, causing failures on specific samples [4, 33]. Therefore, stronger cross-modal alignment and robust feature extraction are crucial to tackle challenges like data quality and noise interference.

2.3 LLM-Enhanced Fake News Detection

Recent studies [6, 7, 46] highlight the strong reasoning capabilities of large language models (LLMs), demonstrating their effectiveness in fake news detection [52, 53, 58]. LLMs have expanded into the visual domain, processing multimodal information. LVLMs, such as LLaVA [28] and MiniGPT4 [62], pioneered image-text alignment and visual instruction fine-tuning, improving multimodal performance, including fake news detection [30, 40]. For example, **NLIN** [60] improves fake news detection by unifying multimodal content and fine-tuning LLMs with external knowledge.

However, LVLM-based models often miss image details and spatial relationships, limiting their ability to detect cross-modal inconsistencies in news. Moreover, fine-tuned LVLMs are prone to hallucinations—misrecognizing unseen objects as familiar ones from the training set—which impacts the stability and effectiveness of cross-modal alignment in fake news detection [8, 17].

3 Methodology

This section presents CMA-VRP for detecting multimodal fake news, as shown in Figure 3. We first construct unimodal graphs for text and images, extracting visual reasoning features (VRFs) by inputting text and multimodal data into the LLM to extract text and visual entities. To capture fine-grained cross-modal interactions, we build a multimodal graph through entity matching and apply intra- and inter-graph contrastive learning for semantic alignment. Cross-modal graph fusion then provides a semantically consistent representation. To obtain robust, modality-invariant features and leverage unpaired samples in fake news, we use visual-semantic cycle alignment, project features into a shared subspace, and optimize through weighted aggregation. Finally, the combined semantically consistent and modality-invariant representations are used for fake news detection. The model consists of three parts: entity-driven graph and visual reasoning initialization, cross-modal graph alignment and fusion, and adversarial cycle-consistent invariant learning.

3.1 Entity-Driven Graph and Visual Reasoning Initialization

This module constructs unimodal graph representations and extracts visual reasoning features, forming the basis for multimodal alignment and reasoning. As shown in the left part of Figure 3, it consists of two components: Unimodal graph construction, models fine-grained semantic relationships between text and images via entities. Truth-driven CoT-based visual reasoning, leverages LLMs and LVLMs with visual prompts to extract deep reasoning features, establishing a semantic foundation for multimodal fusion and alignment.

3.1.1 Unimodal Graph Construction. Given a news pair (I, T), we construct a textual graph $\mathcal{G}_t = (\mathcal{V}_T, \mathcal{E}_T)$ and a visual graph $\mathcal{G}_v = (\mathcal{V}_I, \mathcal{E}_I)$ to model the syntactic and visual structures of the content, as shown in Figure 2 (a), where $\mathcal{V}$, and $\mathcal{E}$ denote nodes and edges, respectively. The textual graph, generated via dependency parsers (e.g., Stanford NLP [34] and TextSmart NLP tools [29]), captures structural relationships with nodes ($\mathcal{V}_t$) representing syntactic units and edges ($\mathcal{E}_t$) their connections. The adjacency matrix (A_t) uses 0 or 1 to represent the presence or absence of edges between nodes, textual embeddings (X_t) are extracted using the BERT [9], and a trainable edge transformation matrix (Z_t) refines semantic dependencies.

The visual graph, based on a scene graph generation model [44] (Mask-RCNN [14] as the backbone), identifies key objects and their relationships (e.g., holding and wearing). The top k objects, ranked by classification scores, form valid visual entities ($\mathcal{V}_v$), while spatial/semantic relations form edges ($\mathcal{E}_v$), encoded in the adjacency mask matrix (A_v). Nodes ($\mathcal{V}_v$) and edges ($\mathcal{E}_v$) are represented by feature vectors (X_v and Z_v), which are derived from Mask-RCNN outputs. Specifically, we encode the entire image as a global node (v_g), linking it to all entities to provide global visual semantics ensuring holistic context understanding. Rosetta OCR [2] extracts textual elements from images, such as slogans, which are then

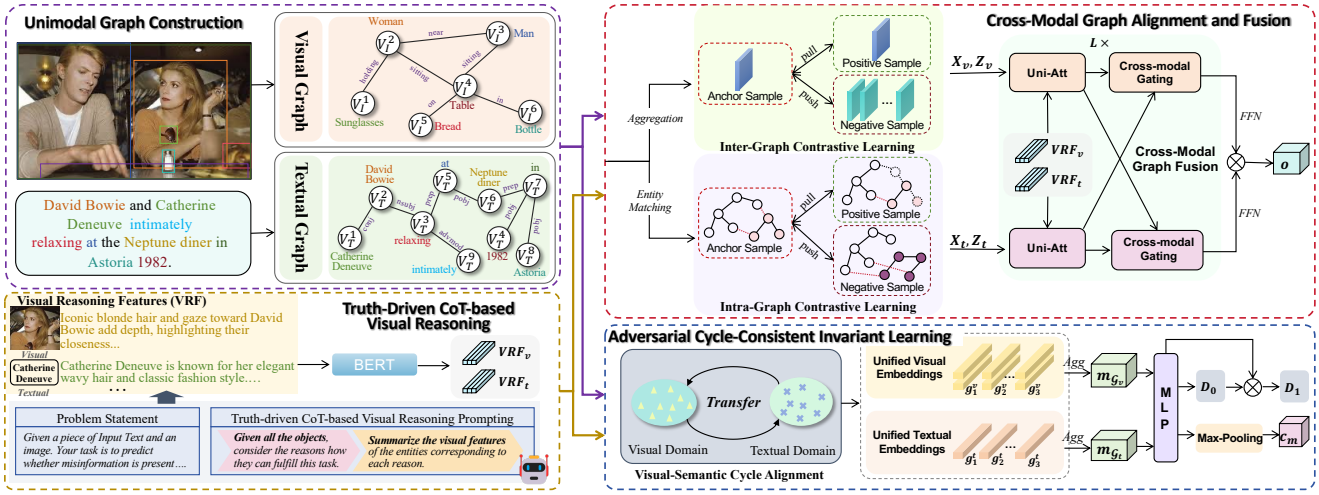


Figure 3: The overall architecture of CMA-VRP.

parsed similarly to the above, encoded as supplementary nodes (v_{ocr}) in the visual graph and linked to visual entities ($\mathcal{V}_v$) through spatial/semantic edges.

3.1.2 Truth-Driven CoT-based Visual Reasoning. By analyzing the visual features of each object, we provide reasoning support related to spatial relationships, behavioral dynamics, and knowledge context, thereby offering visual reasoning information for fake news detection. To prevent cross-modal interference, we employ dual-layer Visual Reasoning Prompting (VRP) to extract visual reasoning features (VRF) from the textual and image entities in the graph construction process for LLMs and LVLMs, both without any fine-tuning. For example, Mistral [18] processes textual entities, while LLaVA [27] handles visual entities. Given object examples, we prompt the language model to generate reasoning, as illustrated in Figure 2 (b).

***Prompt**: For each object, consider the rationale for how its visual features justify whether the claim contains misinformation. **Output**: (Textual Entity) Protester: It is necessary to determine whether the scene depicts a protest and whether the individuals' actions align with the description of a "peaceful" protest. Bolton: ... (Visual Entity) Man_0: The man appears to be a protester. If the protester's demeanor and actions contradict the description, it points to a discrepancy...*

Through this prompting, we generate reasoning texts describing entities' visual characteristics. We then further prompt to summarize each object's relevant visual features as follows:

***Prompt**: Summarize the corresponding visual features of each object for each rationale. **Output**: (Textual Entity) { The protesters should appear calm and non-confrontational, with no signs of aggression or violence. }, {...} (Visual Entity) { Intense facial expression, tightly restrained by police...}, {...}, {...}*

We then obtain visual reasoning texts for N_t textual entities and N_v visual entities, which are encoded through the BERT [9] to obtain the visual reasoning features VRF_t and VRF_v.

3.2 Cross-Modal Graph Alignment and Fusion

This section presents cross-modal graph alignment and graph fusion. Inspired by contrastive learning's role in fake news detection, we innovatively propose fine-grained graph alignment in news detection by leveraging contrastive learning to align nodes and edges in textual and visual graphs, ensuring semantic consistency and reducing modality biases. Graph fusion integrates textual and visual information via aligned graph representations, enhancing cross-modal interactions for a unified representation.

3.2.1 Graph Constrative Learning. We enhance cross-modal alignment and semantic consistency through inter-graph intra-graph and intra-graph contrastive learning.

Inter-Graph Contrastive Learning. We aim to align the semantic representations between textual and visual modalities through inter-graph contrastive learning while capturing semantic consistency within modality-specific graphs. For a batch of N image-text pairs $\{x_i\}_{i=1}^N$, where each pair serves as an anchor sample and includes textual and visual modalities denoted as $x_i = \{s_t^i, s_v^i\}$, the corresponding textual and visual graphs are represented as $\{(\mathcal{G}_t^i, \mathcal{G}_v^i)\}_{i=1}^N$, where i represents the i-th pair. The training batch contains $N^2 - N$ negative samples. Therefore, we can get large volumes of positive examples $\mathcal{X}^+$ and negative examples $\mathcal{X}^-$ for each image-text pair (s_t, s_v). The contrastive loss aligns textual (X_t^m) and visual (X_v^m) graph representations, ensuring non-matching pairs are effectively separated. To mitigate potential semantic ambiguities in cross-modal alignment, we introduce a learnable smoothing parameter $\theta \in [0, 1]$ that adaptively balances strict one-hot alignment and similarity-based soft alignment.

To compute the inter-graph contrastive loss, we first need the textual and visual graph representations. Following [55], we use the READOUT(·) function to aggregate the node representations of each graph:

$$X_v^m = \text{READOUT}^v(\{h_i : v_i \in \mathcal{V}_I\}), \quad X_t^m = \text{READOUT}^t(\{h_i : v_i \in \mathcal{V}_T\}), \tag{1}$$

where h_i represents the node embeddings, and $\mathcal{V}_I$ and $\mathcal{V}_T$ denote the sets of visual and textual nodes, respectively. $X_t^m = f(s_t)$ and $X_v^m = f(s_v)$ denote the fused textual and visual features. We redefine the alignment target by combining the standard one-hot labels Y_{ij} with a similarity-guided soft alignment matrix S_{ij}:

$$S_{ij} = \frac{\exp(\text{sim}(X_t^m[i], X_v^m[j])/\tau)}{\sum_{j'} \exp(\text{sim}(X_t^m[i], X_v^m[j'])/\tau)}, \quad \tilde{Y}_{ij} = \theta \cdot Y_{ij} + (1-\theta) \cdot S_{ij}, \tag{2}$$

where θ is a learnable parameter initialized as 0.5. The inter-graph contrastive loss is then modified as:

$$\mathcal{L}_{inter}=\mathbb{E}\left[-\sum_{i,j}\tilde{Y}_{ij}\log\frac{\exp(\text{sim}(f(s_t[i]),f(s_v[j]))/\tau)}{\sum_{j'}\exp(\text{sim}(f(s_t[i]),f(s_v[j']))/\tau)}\right]. \quad (3)$$

The similarity function sim() measures the cosine similarity between the two representations, and τ is the temperature parameter.

Intra-Graph Contrastive Learning. To enhance the model's cross-modal understanding in fake news detection, we align representations of the same entities across different modalities during the training phase, which helps reduce semantic bias between modalities. First, entity matching is performed to construct the multimodal graph, followed by entity masking to generate positive samples, thereby strengthening local semantic reasoning and ensuring that the model can maintain accurate understanding of news content even when faced with missing entities or ambiguous information. Additionally, negative sample mining strengthens the model's ability to distinguish correct from incorrect modality alignments.

(i) Entity Matching. Following [13, 60], we leverage LVLM's inherent strong visual entity recognition capability to identify visual entities $\mathcal{V}_v$ and verify whether corresponding textual entities exist in the textual graph $\mathcal{G}_t$, linking the textual graph $\mathcal{G}_t$ and the visual graph $\mathcal{G}_v$ to construct the multimodal graph $\mathcal{G}_m = \{\mathcal{V}_m, \mathcal{E}_m\}$. The goal is to utilize consistent cross-modal entities in contrastive learning to optimize the overall encoding representations across different modalities.

(ii) Entity Masking. We apply entity masking to $\mathcal{G}_m$, producing the augmented graph $\mathcal{V}_+(\mathcal{G})$. Entity nodes are masked with probability p_m, their features replaced by random embeddings to simulate removal, guiding the model to focus on remaining key entities and relations while mitigating over-reliance on individual nodes and enhancing global semantic understanding. The augmented graph is: $\mathcal{V}_+(\mathcal{G}) = (M \otimes \mathcal{V}_m, \mathcal{E}_m)$ where M is the masking vector for entity nodes $\mathcal{V}_m$, while $\mathcal{E}_m$ remains unchanged to preserve relational information.

(iii) Negative Mining. To counteract the dominance of one modality, we have implemented a perturbation strategy within our contrastive learning framework to create negative samples. We randomly blend samples of the same modality with opposite labels and randomly connect cross-modal entities to form negative samples. From a sample set $\{x_i\}_{i=1}^N$, where the normal samples include the corresponding text and visual modal denoted as $x_i^a = \{s_t^i, s_v^i\}$ severs as anchor samples, we differentiate between positive pairs (x_i^a, x_i^+) and perturbed negative pairs (x_i^a, x_i^-). Here, x_i^+ is derived from the entity masking process. The latter are generated by using a random extraction function $d(\cdot)$ to mix modalities. This process enhances discrimination and understanding of modality interaction. This is represented as $x_i^- = \{s_t^i, s_v^{d(i)}\}$ or $x_i^- = \{s_t^{d(i)}, s_v^i\}$, with its multimodal graph denoted as $V_-(\mathcal{G})$. Our method pulls anchors and positives closer while pushing negatives apart.

We apply the InfoNCE [36] loss, following SimCLR [3], to enhance pair consistency. The goal is to maximize the cosine similarity for positive pairs while minimizing it for negative pairs through the contrastive loss:

$$\mathcal{L}_{intra}=-\mathbb{E}_{x\sim\mathcal{X}}\left[\log\frac{\exp(sim(g(x^a),g(x^+))/\tau)}{\sum_{x^-\in\mathcal{X}^-}\exp(sim(g(x^a),g(x^-))/\tau)}\right], \quad (4)$$

where sim is the cosine similarity, $g(\cdot)$ denotes the representation after cross-modal graph fusion (see Sec. 3.2.2), and τ is a scaling temperature parameter. Finally, the intra-graph contrastive learning and inter-graph contrastive learning are combined as follows: $\mathcal{L}_{cl} = \mathcal{L}_{intra} + \mathcal{L}_{inter}$.

3.2.2 Cross-modal Graph Fusion. This module (the upper right part of Figure 3) enhances graph-level interactions, improving visual-text integration through fine-grained cross-modal links and rich semantic interactions to reinforce cross-modal consistency and deep semantic understanding of news content. Since some textual words lack direct visual counterparts, we define four generic type nodes: misc, person, location, and organization, linking each textual word to visual nodes detected via generic words through inter-modal edges. Since Graph Convolutional Networks (GCNs) [56] can capture latent entity associations and reduce irrelevant interference, we apply L graph-based multimodal fusion layers to process the multimodal graph, capturing intra- and inter-modal semantic interactions. This encoding refines node states by integrating syntactic dependencies and cross-modal semantics. For intra-modal fusion, Unimodal Attention (Uni-Att) within a Transformer framework enhances text and visual representations, reinforcing syntactic relations and associative links within each modality.

For inter-modal fusion, a cross-modal gating mechanism refines node states by integrating semantic information from cross-modal neighbors, enhancing multimodal comprehension. Visual reasoning features (VRF_t) capture implicit text-image semantic relations, improving alignment. We append VRF_t to textual nodes during fusion to enrich semantic understanding. The intra-modal fusion is expressed as:

$$C_t^{(l)}=\text{Softmax}(A_t\cdot\frac{Q_t^{(l-1)}(K_t^{(l-1)})^\top}{\sqrt{d}})\cdot V_t^{(l-1)}, \quad (5)$$

where the queries, keys, and values for all words are represented by $Q_t^{(l)}, K_t^{(l)}, V_t^{(l)} \in \mathbb{R}^{n\times d}$, defined as: $Q_t^{(l-1)} = W_Q^{(l-1)} \cdot H_t^{(l-1)}$, $K_t^{(l-1)} = W_K^{(l-1)} \cdot H_t^{(l-1)} + \phi(Z_t)$, $V_t^{(l-1)} = W_V^{(l-1)} \cdot H_t^{(l-1)} + \phi(Z_t)$, where ϕ is a learnable function, such as a linear layer, and $W_Q^{(l-1)}$, $W_K^{(l-1)}$, $W_V^{(l-1)}$ are the parameter matrices for the attention mechanism. The initial value is $H_t^{(0)} = X_t$.

For the inter-modal fusion, incorporating the visual reasoning features, the fusion layer is updated as follows:

$$R_{t_u}^{(l)}=C_{t_u}^{(l)}+\sum_{j\in\mathcal{N}_v(u)}\alpha_{i,j}\cdot C_{v_j}^{(l)}+W_{vt}^{(l)}\cdot VRF_t(u), \quad (6)$$

$$\alpha_{i,j}=W_1^{(l)}C_{t_u}^{(l)}+W_2(l)C_{v_j}^{(l)}, \quad (7)$$

where $\mathcal{N}_v(u)$ represents the set of neighboring visual nodes of u, and W_{vt} is a learnable weight for incorporating the visual reasoning features VRF_t.

The final node state $C_t^{(l)}$ for textual nodes, now enhanced with the reasoning features, is obtained using a position-wise feed-forward network (FFN): $H_t^{(l)} = \text{FFN}\left(R_t^{(l)}\right)$. Similarly, the final node state $H_v^{(L)}$ for visual nodes is computed in parallel. After L layers of fusion, the final textual and visual object representations, $H_t^{(L)}$ and $H_v^{(L)}$, are aggregated through the READOUT function to yield $m_{\mathcal{G}_m}$, which results in the semantic consistency representation $o = m_{\mathcal{G}_m}$.

3.3 Adversarial Cycle-Consistent Invariant Learning

To enhance cross-modal alignment robustness and capture deep semantic features, this module (the bottom right part of Figure 3) employs modality-domain transformation learning to map image and text entities into a shared semantic space, uncovering latent associations without explicit alignment, extracting modality-invariant features, and leveraging unpaired samples to improve noise resistance.

3.3.1 Visual-Semantic Cycle Alignment. To fully utilize unmatched entities and learn cross-modal semantic mappings while mitigating semantic divergence from modality-specific noise, we align entity embeddings from textual ($\mathcal{G}_t$) and visual ($\mathcal{G}_v$) graphs via modality-domain transformation. We adopt a cycle-consistent adversarial network [63] with wasserstein distance [38] to reduce domain discrepancies and improve cross-modal consistency. Incorporating visual reasoning features (VRF) into the Wasserstein distance further enhances alignment based on task-specific semantics:

$$W(\mathcal{G}_t,\mathcal{G}_v)=\inf_{\gamma\sim\Pi(\mathcal{G}_t,\mathcal{G}_v)}\mathbb{E}_{(u,w)\sim\gamma}[\,wd(u,w)\cdot|w-u|\,], \tag{8}$$

where $wd(u,w)=\mathrm{sim}(VRF_t(u),VRF_v(w))$ denotes similarity between textual and visual entities.

The discriminator D_s distinguishes node embeddings from different modalities to maintain task-related semantic consistency, while the feature extractor minimizes the Wasserstein distance. The discriminator loss is defined as:

$$\mathcal{L}_{cycle}=-\mathbb{E}_{u\sim\mathcal{G}_t}[D_s(u,VRF_t(u))]+\mathbb{E}_{w\sim\mathcal{G}_v}[D_s(w,VRF_v(w))], \tag{9}$$

where $\mathcal{L}_G^*=-\mathcal{L}_{D_s}^*$. This loss enhances semantic consistency, reduces noise impact, and facilitates modality-invariant feature extraction. After alignment, entity representations are aggregated via the READOUT function to obtain unified modality representations e_v and e_t, capturing refined semantic features for each modality and enhancing robustness in multimodal fake news detection.

3.3.2 Modality-Invariant Feature Extraction. To improve robustness of cross-modal alignment and capture deep semantic features, we propose a dual-discriminator framework to generate modality-invariant feature, where the first discriminator D_0 identifies modality-invariant features within a shared latent space, refining representations for challenging the second discriminator.

Specifically, discriminator D_0 is a binary classifier operating in a shared latent subspace, guiding the projection I: $D_0(x)=\mathrm{softmax}(D_0(I(x)))$. Here, $x=c=I(e)$, where e is the aligned modality embedding, and I is the invariant projection layer, and c is the projected modality-invariant feature. Assuming D_0 is optimally functioning, $D_0^i(x)$ indicates the probability of modality i, influencing the modality-invariant weighting as $w_i(x)=1-D_0^i(x)$.

The adversarial framework, following [49], minimizes the Jensen-Shannon divergence (JSD) through a minimax game, with the second discriminator D_1 aiming to blend modality features optimally. The weighted adversarial objective is:

$$\begin{aligned}\min_I\max_{D_1}\mathcal{L}_{D_1}=&\,\mathbb{E}_{e\sim pe_t}[\,w_t(x)\log(D_1(I(e)))]\\&+\mathbb{E}_{e\sim pe_v}[\,w_v(x)\log(1-D_1(I(e)))],\end{aligned} \tag{10}$$

where $w(x)$ is a modality-invariant weighting, independent of D_1, pe_t and pe_v are distributions of two modalities. For fixed I and D_0, the optimal output for D_1^i:

$$D_1^{i*}(x)=\frac{w_i(x)pe_i(x)}{w_t(x)pe_t(x)+w_v(x)pe_v(x)}. \tag{11}$$

The optimal D_1 condition leads to the minimax game resolving as:

$$\mathcal{L}_{D_1}=-\log(10)+2\cdot\mathrm{JSD}(w_t(x)\cdot pe_t(x)||w_v(x)\cdot pe_v(x)). \tag{12}$$

The Jensen-Shannon divergence (JSD) reaches its global minimum when the distributions are equal, $w_t(x)pe_t(x)=w_v(x)pe_v(x)$, which aligns precisely with the goal of cross-modal adversarial learning [12]. Given $w_i(x)=w_i(I(e_i))$ and $pe_i=I(e_i)=c_i$, we obtain modality-invariant representation c_m via max-pooling over weighted modality-invariant features: $c_m=max\{w_tc_t,w_vc_v\}$. The loss of the entire module is: $\mathcal{L}_{adv}=\mathcal{L}_{D_0}+\mathcal{L}_{D_1}+\mathcal{L}_{cycle}$, combining both discriminators and cycle consistency for robust cross-modal alignment and extraction of modality-invariant features.

The final news representation $\hat{x}$ concatenates multimodal graph features (o) from contrastive learning and modality-invariant features (c_m): $\hat{x}=o\oplus c_m$, where $\oplus$ denotes concatenation. A fully connected layer classifies this representation:

$$\hat{y}=\mathrm{softmax}(w\hat{x}+b),\quad \mathcal{L}_y=-y\log(\hat{y})-(1-y)\log(1-\hat{y}), \tag{13}$$

where $\hat{y}$ is the predicted label distribution, and w,b are trainable parameters. The model optimizes cross-entropy loss ($\mathcal{L}_y$), with total loss: $\mathcal{L}_{total}=\mathcal{L}_y+\lambda\mathcal{L}_{cl}+\beta\mathcal{L}_{adv}$, where λ and β as hyperparameters managing the balance between these loss components.

4 Experiments

Datasets. Our research evaluates our method on three representative real-world datasets: Twitter, Pheme, and Weibo. The Twitter dataset, used in MediaEval's Verifying Multimedia Use task [1], contains 7,898 real and 6,026 fake tweets for training, and 1,972 real and 3,830 fake tweets for testing. The Pheme dataset, for social media news verification, includes 1,972 real and 3,830 fake news samples from five breaking stories, with 3,670 images [41]. The Weibo data set, for the detection of fake news, has 6,840 real tweets and 5,007 fake tweets in training, and 1,000 fake and 996 real news in testing [19].

Baselines. We evaluate our method and baselines: **BERT** [9], a transformer-based model; **ResNet34** [42], a convolutional network for image classification; **MVAE** [20], a multimodal variational autoencoder for shared text-image representations; **CFFN** [22], a network aligning visual semantics and text for fake news detection via word-region pair consistency; **InfoSurgeon** [11], detecting fake news by cross-media entity, relation, and event consistency in knowledge graphs; **Event-Radar** [33], detecting fake news via event-level inconsistencies and multi-view feature integration; **COOLANT** [51], using contrastive learning and attention for multimodal fake news detection. For LLM-based methods, we assess **LLaVA** [27], a strong LVLM based on vision instruction tuning; **LLaVA + SPT**, which integrates soft prompt tuning [21]; and **NLIN** [60], unifying multimodal news into a natural language space using an encoder-decoder with prompts for detection. These models include unimodal models (BERT, ResNet34), cross-modal alignment methods (CFFN, MVAE, CAFE, InfoSurgeon, COOLANT), LVLM-based models (LLaVA), and fine-tuned LVLMs (LLaVA + SPT, NLIN).

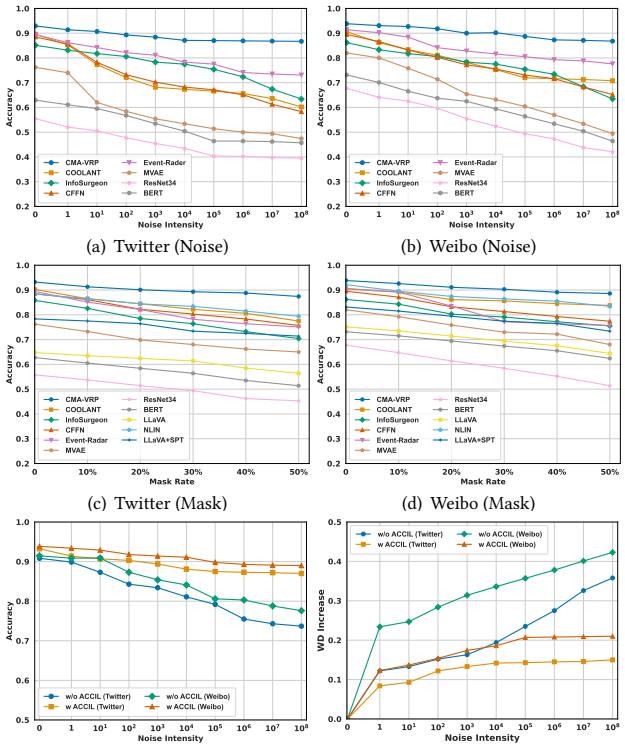
Table 1: Experimental results on Twitter, Pheme, and Weibo datasets. And with * indicates the significance test over Event-Radar and COOLANT presents a statistically significant improvement (with $p < 0.05$).

	Method	Accuracy	Fake News			Real News		
			Precision	Recall	F1-score	Precision	Recall	F1-score
Twitter	BERT	0.629	0.801	0.489	0.608	0.575	0.750	0.651
	ResNet34	0.550	0.535	0.534	0.534	0.513	0.513	0.513
	MVAE	0.757	0.831	0.661	0.772	0.738	0.713	0.772
	CFFN	0.886	0.898	0.884	0.882	0.887	0.882	0.889
	Event-Radar	0.902	0.892	0.903	0.901	0.932	0.883	0.910
	InfoSurgeon	0.855	0.851	0.849	0.850	0.859	0.858	0.859
	COOLANT	0.908	0.904	0.902	0.903	0.912	0.911	0.912
	LLaVA	0.641	0.687	0.584	0.680	0.613	0.632	0.624
	LLaVA+SPT	0.783	0.813	0.693	0.790	0.743	0.785	0.774
	NLIN	0.883	0.916	0.796	0.893	0.875	0.888	0.867
	CMA-VRP	**0.930***	**0.938**	**0.923***	**0.926***	**0.934***	**0.945**	**0.932***
Pheme	BERT	0.713	0.875	0.563	0.682	0.649	0.824	0.725
	ResNet34	0.649	0.751	0.439	0.568	0.535	0.710	0.611
	MVAE	0.776	0.801	0.749	0.795	0.745	0.746	0.798
	CFFN	0.869	0.872	0.877	0.854	0.872	0.869	0.874
	Event-Radar	0.901	**0.920**	0.822	0.831	0.905	0.904	0.902
	InfoSurgeon	0.859	0.799	0.780	0.829	0.833	0.823	0.847
	COOLANT	0.883	0.860	0.903	0.881	0.904	0.864	0.882
	LLaVA	0.664	0.734	0.623	0.698	0.638	0.617	0.627
	LLaVA+SPT	0.792	0.796	0.793	0.810	0.793	0.787	0.785
	NLIN	0.903	0.887	0.904	0.883	0.863	0.868	0.875
	CMA-VRP	**0.926***	0.913	**0.916***	**0.927***	**0.922***	**0.927**	**0.926***
Weibo	BERT	0.731	0.730	0.731	0.731	0.730	0.731	0.731
	ResNet34	0.675	0.682	0.665	0.664	0.685	0.683	0.679
	MVAE	0.820	0.855	0.780	0.816	0.791	0.861	0.825
	CFFN	0.894	0.876	0.817	0.841	0.902	0.915	0.899
	Event-Radar	0.910	0.881	0.862	0.891	0.915	0.914	0.910
	InfoSurgeon	0.866	0.795	0.776	0.785	0.861	0.868	0.863
	COOLANT	0.915	0.909	0.926	0.914	0.912	**0.924**	0.911
	LLaVA	0.750	0.785	0.713	0.734	0.720	0.771	0.719
	LLaVA+SPT	0.833	0.854	0.795	0.839	0.813	0.832	0.826
	NLIN	0.911	0.913	0.914	0.904	0.907	0.917	0.913
	CMA-VRP	**0.938***	**0.933**	**0.942***	**0.941***	**0.929***	0.906	**0.938***

Sentences were parsed with spaCy (*en_core_web_trf*); text initialized with BERT-based-Uncased [48]; visual features from scene graph generation [44]; visual reasoning from Mistral-7B [18] and LLaVA-1.6-13B [27]. Training: learning rate 0.001 (decay 0.5), early stopping after 5/50 epochs; contrastive $p_m = 0.2$; fusion: $L = 2$, $\lambda = 0.5$, $\beta = 0.1$.

4.1 Results and Analysis

Overall Performance. The performance of baseline models and our proposed CMA-VRP is shown in Table 1, demonstrating CMA-VRP's superior performance. Observations indicate that BERT outperforms ResNet34 in news verification, highlighting the importance of text processing, while deep cross-modal methods like InfoSurgeon, CFFN, and COOLANT generally perform better by integrating multimodal features. CFFN and Event-Radar benefit from capturing fine-grained multimodal cues, leading to better performance, all surpassing MVAE, likely due to MVAE's unsupervised learning, which lacks explicit supervision for deep cross-modal consistency. COOLANT ensures global cross-modal consistency via contrastive learning and modality alignment, but it may struggles with local semantic differences, limiting its detection capability when text-image associations are implicit or indirect. Furthermore, despite improvements, fine-tuned LVLM methods less effective than CMA-VRP likely due to insensitivity to fine-grained variations and constrained generalization from large-model fine-tuning.



(a) Twitter (Noise) (b) Weibo (Noise)

(c) Twitter (Mask) (d) Weibo (Mask)

(e) Impact of ACCIL on Accuracy (f) Impact of ACCIL on WD Increase

Figure 4: Impact of Gaussian noise (a, b), mask ratios (c, d), and noisy conditions on ACCIC variants (e, f).

Our approach excels may due to: (1) Integrating graph modeling and visual reasoning features to capture deep cross-modal semantics. Specifically, through fine-grained contrastive learning, we minimize similarity differences and maximize representational diversity within the same entities or events. (2) Structured graph fusion, modeling entity interactions while preserving spatial and relational semantics, effectively detecting misleading multimodal news by identifying superficially aligned yet semantically contradictory text-image pairs. (3) Modality-domain transformation mitigates dependence on explicitly paired data by aligning related cross-modal entities and descriptions in a shared semantic space, uncovering latent associations in noisy or unaligned data. This enhances cross-modal semantic consistency, enabling inference of implicit relationships and improving detection accuracy.

Robustness Testing. To evaluate CMA-VRP on varying-quality news samples, we injected Gaussian noise into multimodal entity representations [32, 33], simulating information degradation (e.g., noisy text, blurred images). As shown in Figure 4 (a) and (b), we tested accuracy after adding noise to each modality and averaged the results to evaluate the performance of each model. While all models degrade, CMA-VRP shows the smallest accuracy drop and highest stability. We also tested text-image occlusion to assess inference from partial cues. Specifically, we extracted 500 samples from the test set, occluded the text and image regions, and manually filtered the samples to select those with consistent labels before and after occlusion for the robustness test. As shown in Figure 4 (c) and (d), CMA-VRP outperforms both representation learning-based methods and LLM-based methods. This robustness is likely due

Table 2: Ablation study on the architecture design of CMA-VRP across three datasets.

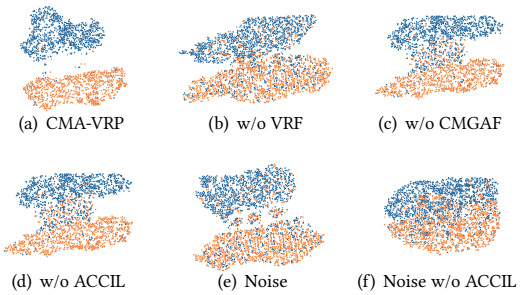
Model	Twitter		Pheme		Weibo	
	Accuracy	F1 score	Accuracy	F1 score	Accuracy	F1 score
CMA-VRP	**0.930**	**0.929**	**0.926**	**0.926**	**0.938**	**0.940**
-w/o VRF	0.905	0.902	0.902	0.900	0.912	0.911
-w/o GCL_{inter}	0.914	0.913	0.908	0.906	0.920	0.918
-w/o GCL_{intra}	0.920	0.921	0.910	0.911	0.916	0.919
-w/o CMGAF	0.895	0.895	0.887	0.894	0.896	0.893
-w/o ACCIL	0.908	0.901	0.896	0.898	0.914	0.909
-w/o VSCA	0.915	0.917	0.905	0.908	0.918	0.920

to CMA-VRP's cross-modal graph fusion, which enhances entity-relation modeling and supplements semantic reasoning with visual features, allowing the model to infer potential logical consistency from remaining information even when part of the modality is missing, thereby reducing misclassification.

To investigate CMA-VRP's robustness to noise and the role of Adversarial Cycle-Consistent Invariant Learning (ACCIL), we analyzed wasserstein distance (WD) across noise level (Figure 4 (e) and (f)). In general, a smaller WD indicates higher distribution overlap, reflecting better cross-modal alignment under noise. Compared to the full model, the w/o ACCIL variant exhibits sharp WD increases and instability under stronger noise, with a steep accuracy drop. This suggests CMA-VRP mitigates noise-induced semantic drift via modality-invariant feature learning, while its adversarial training mechanism enhances cross-modal alignment stability, ensuring high detection accuracy in noisy environments.

Ablation Analysis. We evaluate each component's contribution in CMA-VRP by removing them individually. Variants include w/o VRF (visual reasoning features), w/o GCLinter (inter-graph contrastive learning), w/o GCLintra (intra-graph contrastive learning), w/o CMGAF (cross-modal graph alignment and fusion), w/o ACCIL (adversarial cycle-consistent invariant learning), and w/o VSCA (visual-semantic cycle alignment).

Experimental results indicate that the cross-modal graph alignment and fusion module is crucial to model performance. The observed performance degradation after its removal suggests that modality consistency feature learning plays a key role in cross-modal representation alignment. Both inter-graph and intra-graph contrastive learning improve performance, likely due to their ability to enhance cross-modal semantic understanding, effectively capturing inconsistencies in fake news and allowing the model to better



(a) CMA-VRP (b) w/o VRF (c) w/o CMGAF

(d) w/o ACCIL (e) Noise (f) Noise w/o ACCIL

Figure 5: T-SNE visualizations display the prediction results of CMA-VRP on the Weibo test set.

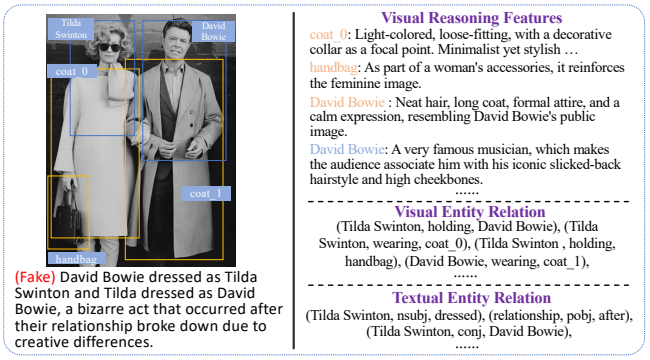


Figure 6: Case Study. Orange represents visual entities, and blue represents textual entities.

distinguish real from fake news. When removing Visual Reasoning Features (VRF), the model still outperforms most baseline methods but experiences a significant performance drop, indicating that the deep visual-semantic information provided by VRF is crucial for enhancing cross-modal reasoning.

4.2 Qualitative Analysis.

We further employ t-SNE [47] to visualize the features before the classifier (Figure 5) for CMA-VRP and its five variants on the Weibo test set. CMA-VRP achieves clear class boundaries, effectively separating real and fake news. Removing cross-modal graph alignment and fusion (w/o CMGAF) blurs these boundaries, highlighting the importance of cross-modal consistency features. When Gaussian noise (10^6 intensity) is introduced, results without adversarial cycle-consistent invariant learning (w/o ACCIL) show decreased robustness, while ACCIL preserves class differentiation under noise. Incorporating visual reasoning features (VRF) further enhances cross-modal understanding and inter-class separability, improving classification.

5 Case Study

Our model's performance in specific scenarios is analyzed through visual reasoning features, textual and visual entity relations in news classification. As shown in Figure 6, cross-modal graph modeling captures fine-grained entity relationships, uncovering subtle inconsistencies like (Tilda Swinton, holding hands, David Bowie), where the contradiction with the textual claim of a broken relationship enhances the model's judgments. Accurate classification highlights the model's sensitivity to fine-grained consistency.

6 Conclusion

We propose Cross-Modal Alignment with Visual Reasoning Prompting (CMA-VRP) to enhance cross-modal alignment and robustnessin multimodal fake news detection. Graph-based semantic alignment with visual reasoning captures deep text-image associations, improving comprehension and reasoning. Adversarial alignment and modality-invariant feature extraction further suppress noise and enhance stability by reinforcing semantic consistency across modalities. Experiments demonstrate that CMA-VRP achieves superior performance and robustness in fake news detection.

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